

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA)
Semester-I University Examination 2009-10
CA703 – Fundamentals of Operating Systems and Scripting

Date: 28/12/2009, Monday

Time: 1:00 p.m. to 4:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicate full marks.
2. Make suitable assumptions wherever necessary & mention them clearly.
3. Answer to the two sections must be written in the separate answer books.
4. Answer to different questions should start on new page.

SECTION – I

Q-1 Do as directed.

- (a) Define Operating System and explain how it provides user interface. [03]
- (b) Write the DOS commands to perform following operations [05]
- Unhide a file named Exam.txt located in MCA directory.
 - Copy the lines of file File1.txt having string "Exam" in file File2.txt.
 - Display all files on drive D: whose 2nd letter is 'C'.
 - Change prompt to current date followed by greater than sign.
 - Display only directory name and sort them in descending order of date.
- (c) What will the following unix commands do? [02]
- tput smso; echo Good Afternoon ; tput rmso
 - who | wc -l

Q-2

- (a) Explain the importance of Cache Memory with its basic design characteristics. [05]
- (b) Explain attrib and xcopy commands. [05]
- (c) Explain Absolute path and Relative path under unix with suitable example. [05]

OR

Q-2

- (a) What is Batch file? In how many ways you can call a batch file ? Also explain **any three** commands used in a batch file with example. [05]
- (b) Explain cmp and comm command with suitable example. [03]
- (c) Explain Time Sharing System. [05]
- (d) Explain uname command with any two options. [02]

Q-3

- (a) Explain the Directory Structure of Unix [05]
- (b) Explain Environment Variables used in Unix. [05]

OR

Q-3

- (a) Explain the concept of Redirection with suitable example. [05]
- (b) Explain /s command with options -l, -R and -F in detail. [05]

P.T.O

SECTION - II

Q-4 Do as directed

- (a) Write command to delete file *chapter** from current directory. [01]
(b) Write the output of command *echo 'My Gross Salary is \$8000'*. [01]
(c) Write command to display home directory in Unix. [01]
(d) What will the following commands do? [04]
1. *cmp file[12]*
2. *tr -s `` < file1 > file2*
3. *set `date`; echo \$1*
4. *tail -3 file1 | head -1*
(e) Explain the meaning of *\$?*, *\$#* and *\$**. [03]

Q-5

- (a) Explain *grep* command and show how you can form regular expressions. [05]
(b) List different modes supported in *vi* editor. Explain commands of any one mode in detail. [05]
(c) How can you change permissions of a file? Show it with example. [05]

OR

Q-5

- (a) How can you perform cursor or screen movement in *emacs* editor. [03]
(b) What do you mean by *inode*? Explain *Hard link* and *Symbolic link* with suitable example. [05]
(c) Explain *cut* command with suitable example. [05]
(d) Give commands to create a file, save a file and exit from *pico* editor. [02]

Q-6

- (a) Write a shell script to calculate the net salary of an employee, where *DA* is 48% of *basic*, *HRA* is 7% of *basic*, *TA* is 5% of *basic* and *PF* is 8% of *basic*. [05]
(b) Write commands in Unix to perform following operations: [05]
• List hidden files of current directory.
• List only directory names in current directory.
• List file names with its size in bytes.
• Delete space from a file *file1*, redirect it to another file *file2* and display contents of *file2*.
• Display message *Today's date is : dd/mm/yy*
Example :- *Today's date is : 28/12/09*

OR

Q-6

- (a) Write a shell script to display the following menu, [06]
1) Display Contents of file.
2) Copy a File.
3) Rename a File.
4) Delete a File.
5) Exit.
(b) Write a shell script to find the sum of *n* numbers passed by command line argument. [04]